

Improvement of Transparency & Stability in Teleoperation using Passivity Control

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5. IRIS lab, KAIST CEE

Contents

I. Hardware Architecture

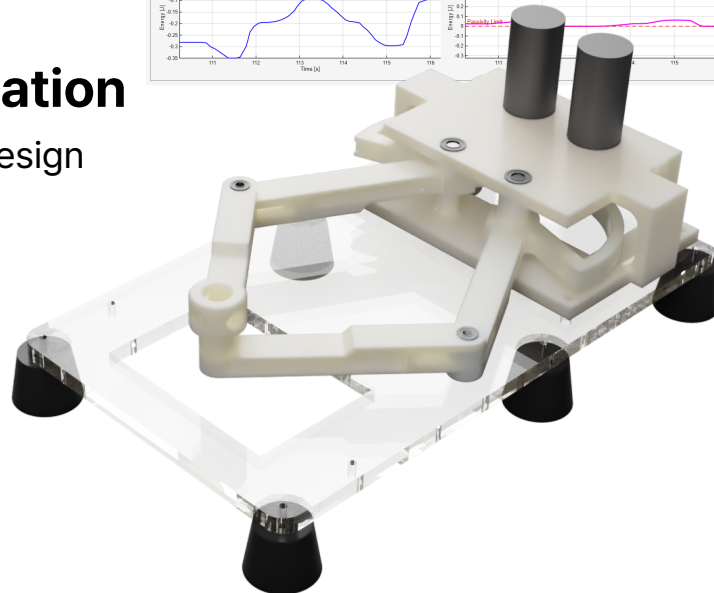
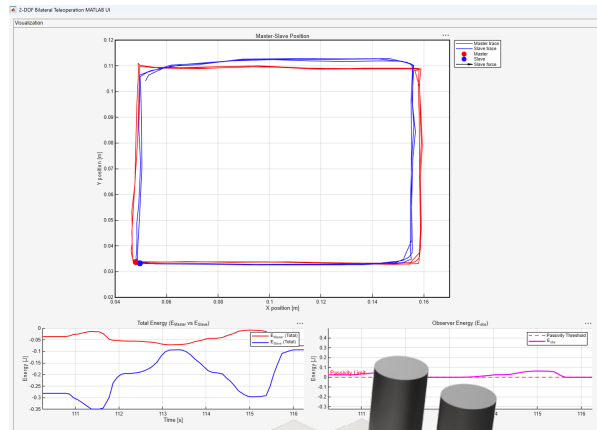
- **Mechanical stability:** Structural modification

II. Groundwork for the Improved Teleoperation

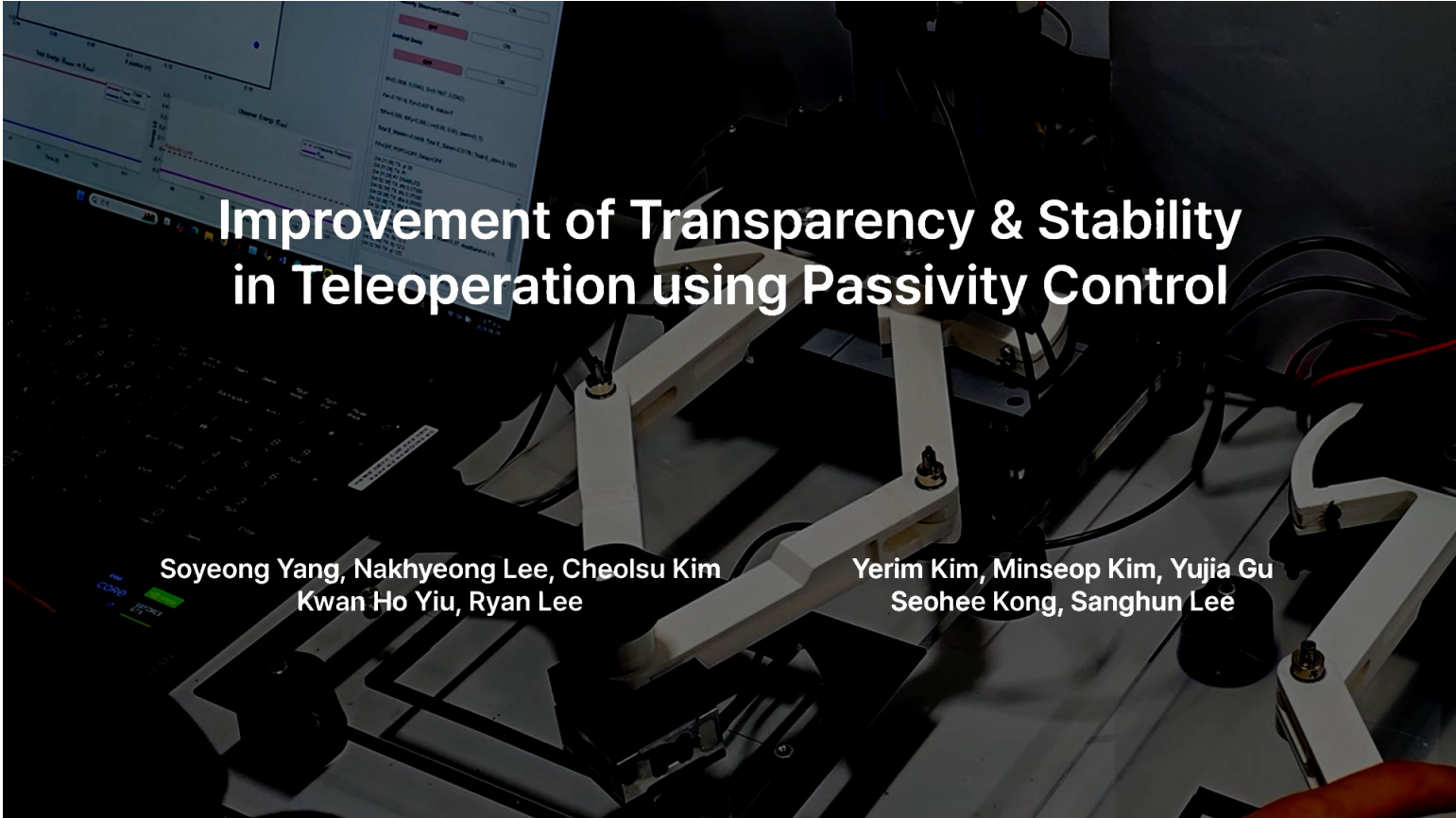
- **Stability & Transparency** focused teleoperation system design

III. Improvements on the Teleoperation

- **Passivity observer**, Effects of the time delay, etc...



FINAL RESULTS



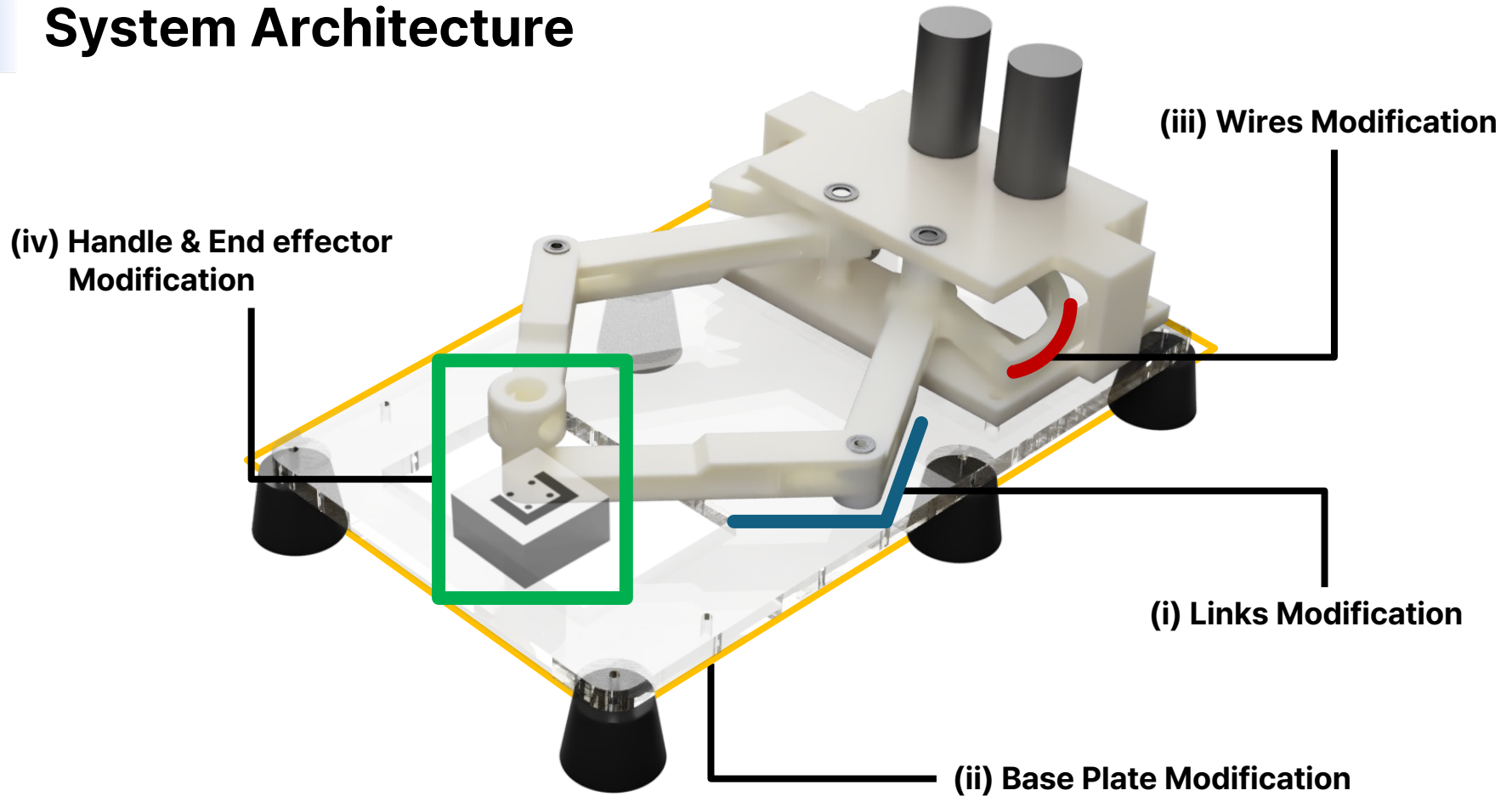
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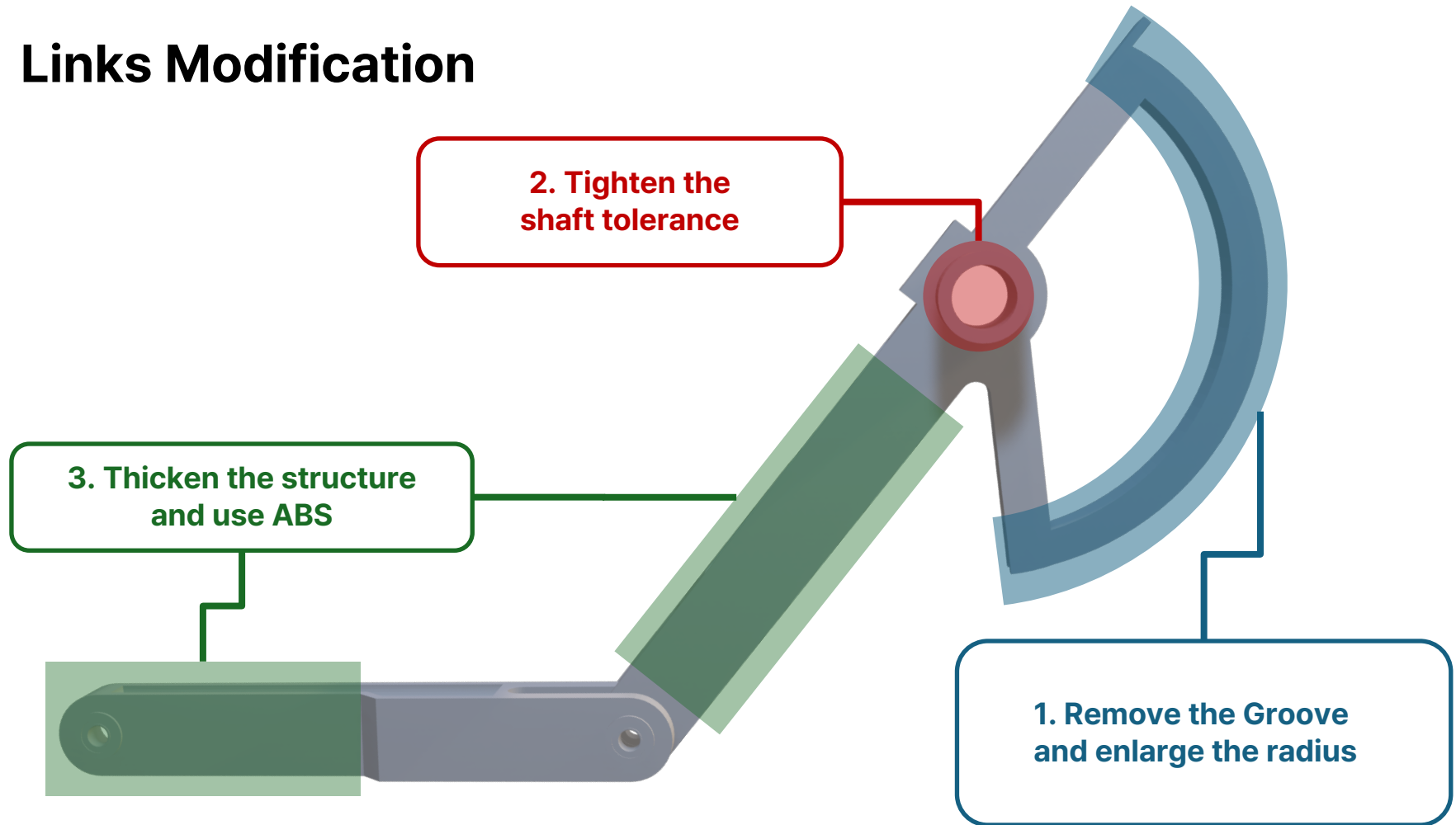
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1. Hardware Architecture

System Architecture

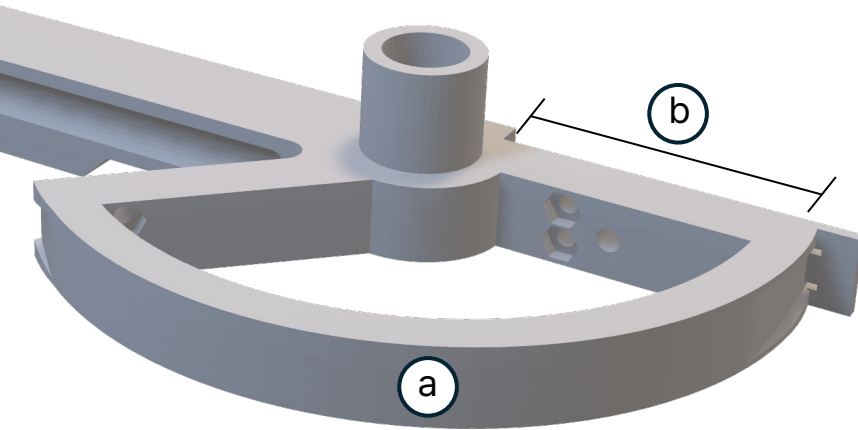


Links Modification



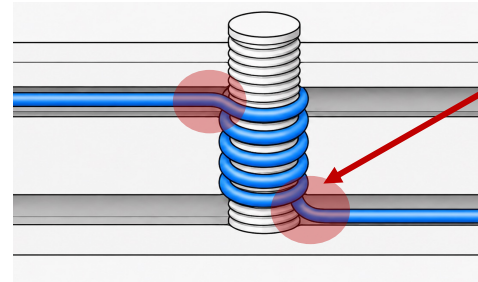
Links Modification

1. Remove the Groove and enlarge the radius



Force equivalent of
lifting **2.1 - 4.2 kg**
weight →

(a)



In the old design, the wire **rubs against the groove wall**. Causing **unexpected friction** to the pulley link's movement.

(b)

Enlarging the pulley link radius (b) to 6.3cm.

While the maximum torque provided by the motor is 0.085Nm, the larger pulleylink design can **enhance the torque to 2.7Nm.**

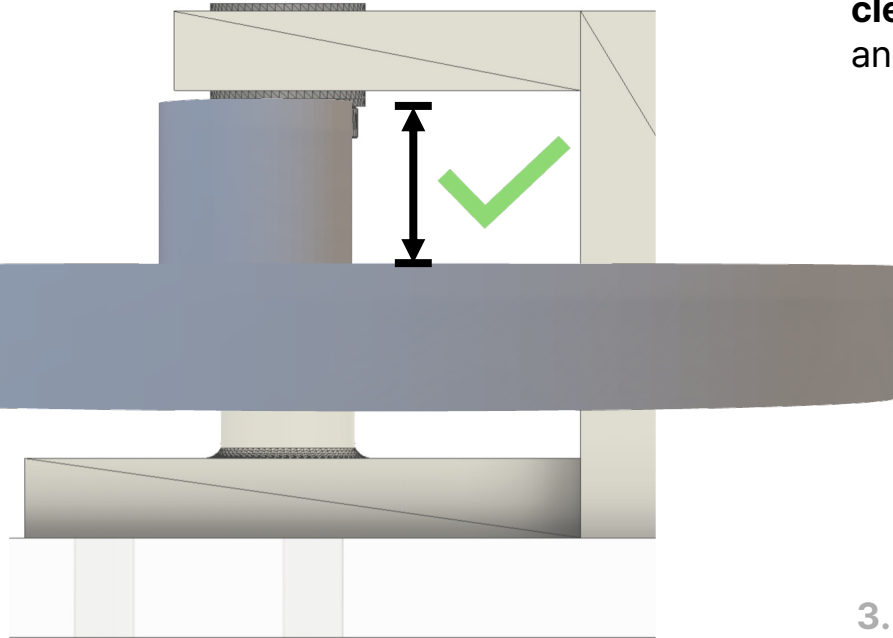
Θ (Angle between links)	Handle Force	Equivalent Mass
30°	20.6 N	2.10 kg
45°	29.2 N	2.98 kg
60°	35.7 N	3.64 kg
90°	41.2 N	4.20 kg

2. Tighten the shaft tolerance

3. Thicken the structure and use ABS

Links Modification

With the additional height:

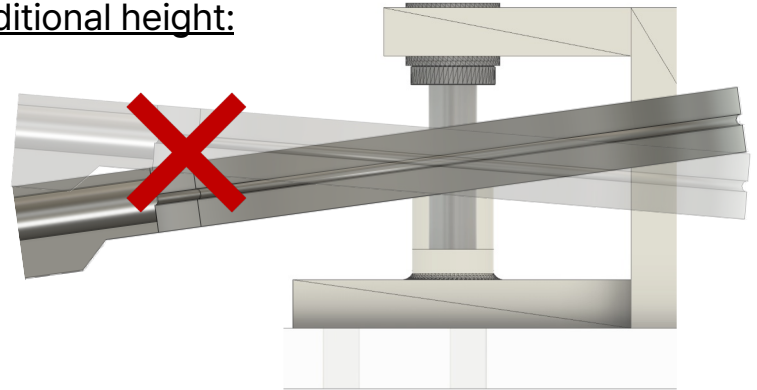


1. Remove the Groove and enlarge the radius

2. Tighten the shaft tolerance

Extrude the pulleylink to **add height** and **eliminate the clearance inside the shaft bore**. This removes free play and **prevents the pulleylink from shaking**

Without the additional height:



3. Thicken the structure and use ABS

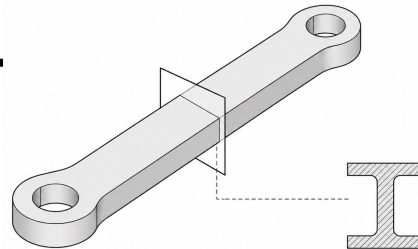
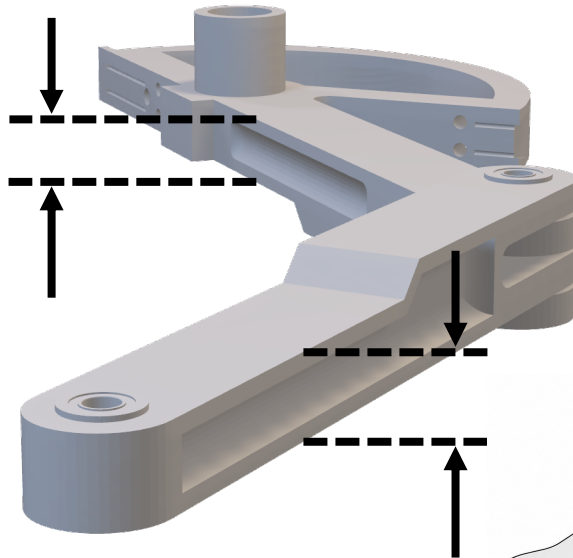
Links Modification

1. Remove the Groove and enlarge the radius
2. Tighten the shaft tolerance

3. Thicken the structure and use ABS

The **structural member highlighted** is thickened from 7.0 mm to 12.6 mm **using the I-beam cross-section**.

In addition, the original material (Bambu Lab PLA Basic) is **replaced with a stiffer, harder ABS** (Stratasys ABS-M30), and **the infill density is increased**.



Material	Modulus
Bambu lab PLA Basic	2.06 GPa
Stratasys ABS-M30	2.30 GPa

$$(EI_x)_{old} = 1.1 \times 10^6 \text{ Nmm}^2$$

$$(EI_x)_{new} = 6.62 \times 10^6 \text{ Nmm}^2$$

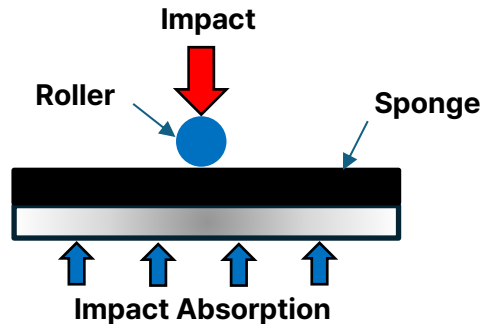

Base Plates Modification

Sponge

- Installed on the Workspace contact area
- Increases the collision time during contact
- Reduces peak impact force & Protect Load cell
- Improve force-feedback stability

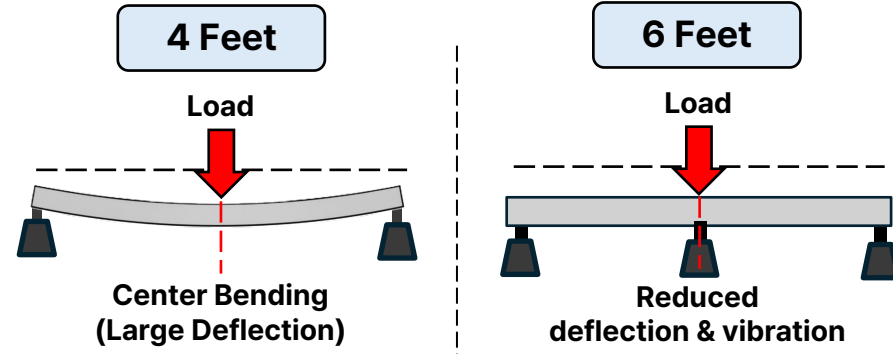
$$F_{avg} = \frac{\Delta p}{\Delta t}$$

$$\Delta t \uparrow \rightarrow F_{impact} \downarrow$$



Auxiliary Rubber Feet

- Initial 4-feet support caused bending at the center
- Added 2 extra feet in the middle
- Reduced center deflection and vibration
- Improve structural stability of the workspace

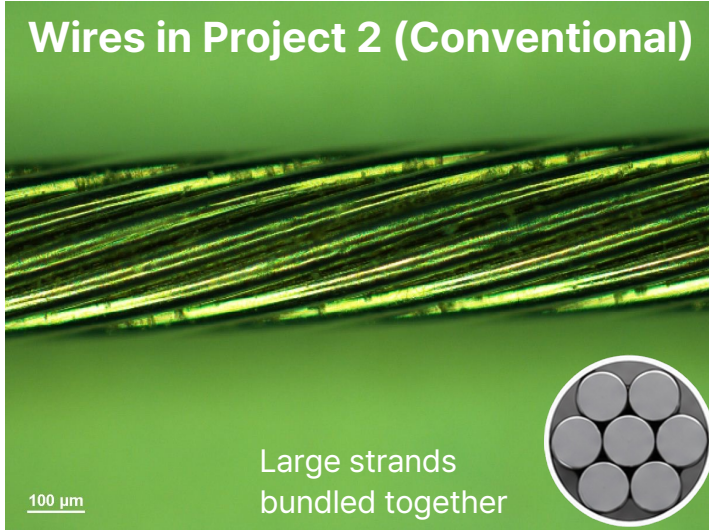


Design Goal : Improve mechanical stability and obtain more reliable haptic interaction before teleoperation experience

Wires Modification

Chosen

Wires in Project 2 (Conventional)



Surface appears
smooth and uniform.
Lower roughness.

**Higher
risk of slip**
✗

Wires in Project 3 (Braided)



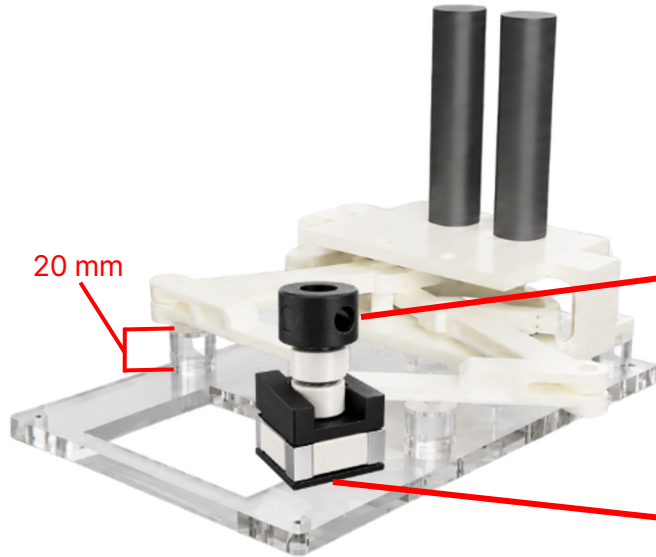
Surface exhibit **greater
waviness and roughness.**
Enhanced mechanical
interlocking with the shaft

**Improved
grip on the
motor shaft**
✓

Handle & End effector Modification

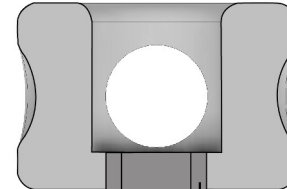
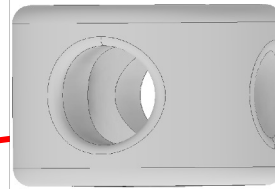
1. 20 mm Height Increase

Preventing contact between the load cell and plate



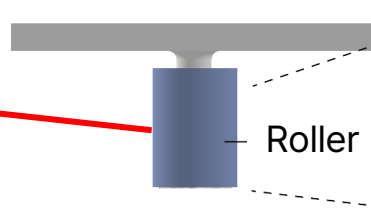
[Follower Device]

2. Handle & End effector (Roller)

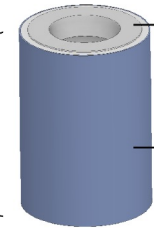


Bearing

Using bearing for a free rotation, **blocking a undesired torque to user**

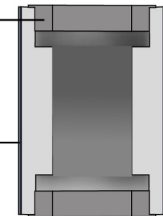


Roller



Bearing

Non-slip
Material



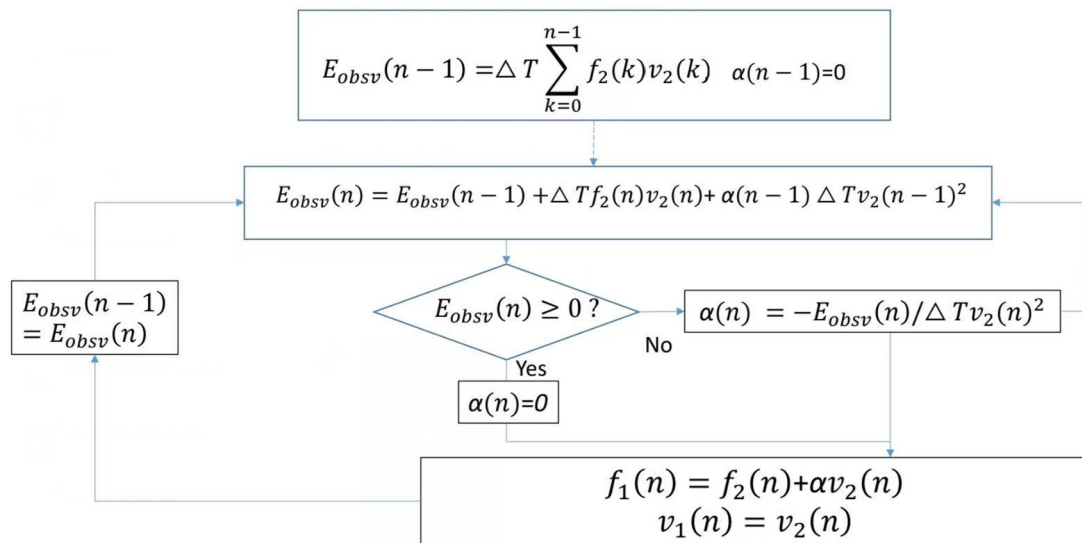
Using roller to decrease friction between end effector and the environment.

2. Groundwork for the Improved Teleoperation

Goal of Improved Teleoperation (Passivity Observer)

Time-Domain Energy Observation and Damping Injection

- **Communication delays and discrete control steps generate non-passive energy, leading to instability.**
- We must observe and dissipate this energy in real-time.



Implication:

Necessity of energy dissipation drives our practical implementation of high-frequency RC filtering.

Theoretical foundation: Achieving Transparency

Ideal Hybrid Matrix in Position-Force Architecture

- Transparency: how perfectly the Master and Slave state variables **map to each other**.
 - Force f and Velocity v
- To achieve perfect transparency, where $f_m = f_s$ and $v_m = v_s$, **we must satisfy** $C_m = -Z_m$

$$\begin{bmatrix} f_m \\ -v_s \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} v_m \\ f_s \end{bmatrix}$$

$$h = \begin{bmatrix} \frac{Z_m + C_m}{Z_s + C_s} & \frac{1}{Z_s + C_s} \\ \frac{-C_s}{Z_s + C_s} & \frac{1}{Z_s + C_s} \end{bmatrix}$$

If C_s goes to infinite

$$h \Rightarrow \begin{bmatrix} Z_m + C_m & 1 \\ -1 & 0 \end{bmatrix}$$

If C_s goes to infinite and $C_m = -Z_m$

$$h \Rightarrow \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

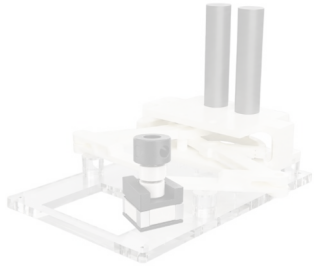
Implication:

Achieving $C_m = -Z_m$ through software alone is highly unstable, **minimizing Z_m through mechanical improvement is essential.**

Source of intransparency and instability

Core Challenges in the Master-Slave System

A. Mechanical Sources



Hardware Limitations

- **Friction and Slack:** cable slipping and loose tension
- **Structural Compliance:** unbalanced collision, tilting

B. Circuit & Signal Sources

PWM (PULSE WIDTH MODULATION) SWITCHING

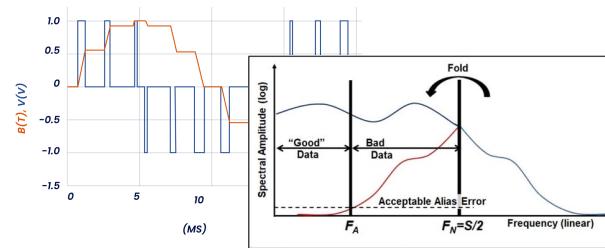
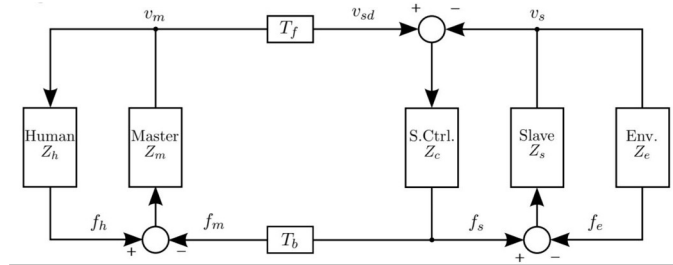


Figure 1... F_A : Acceptable Data Range Without Filtering

Algorithmic Discontinuity

- **High-Frequency crosstalk:** fast-switching motor drivers & clocks.
- **Digital Aliasing:** high-frequency noise sensed as signals in loadcell

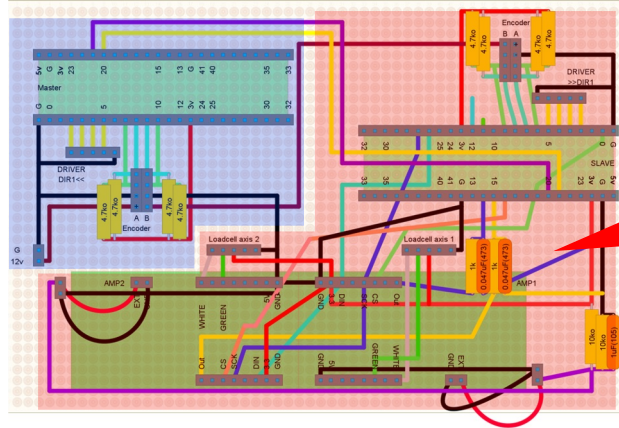
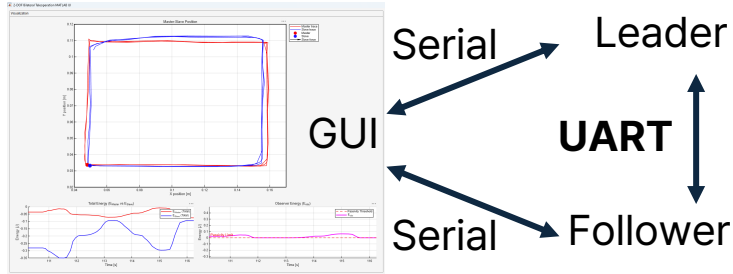
C. Communication Sources



Time Delay

- **Loop Latency:** Routing and blocking functions
- **Rigid Linear Equation:** Sole $F = Kx + Bv$ mapping causes instability

B,C. Communication protocol and circuit design



Loop:
1kHz(1ms)

Crosstalk

PWM	20~100kHz
SPI, UART	~Mbps

- **UART based Direct Microcontroller Link**
 - Leader and Follower Teensy **communicate directly via UART(1kHz)**
 - Reduces PC-side communication **overhead and latency**
 - Less wire is needed than SPI, reduces clock noise and wire-to-wire crosstalk

- **Hardware-Level RC Filter for Anti-Aliasing:**

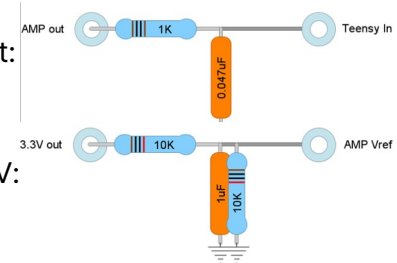
- Implemented the **physical RC low-pass filter** to prevent high-frequency noise in loadcell data from digital system.
- Block >10kHz noise while **keeping low delay (→ Transparency)**

Fast

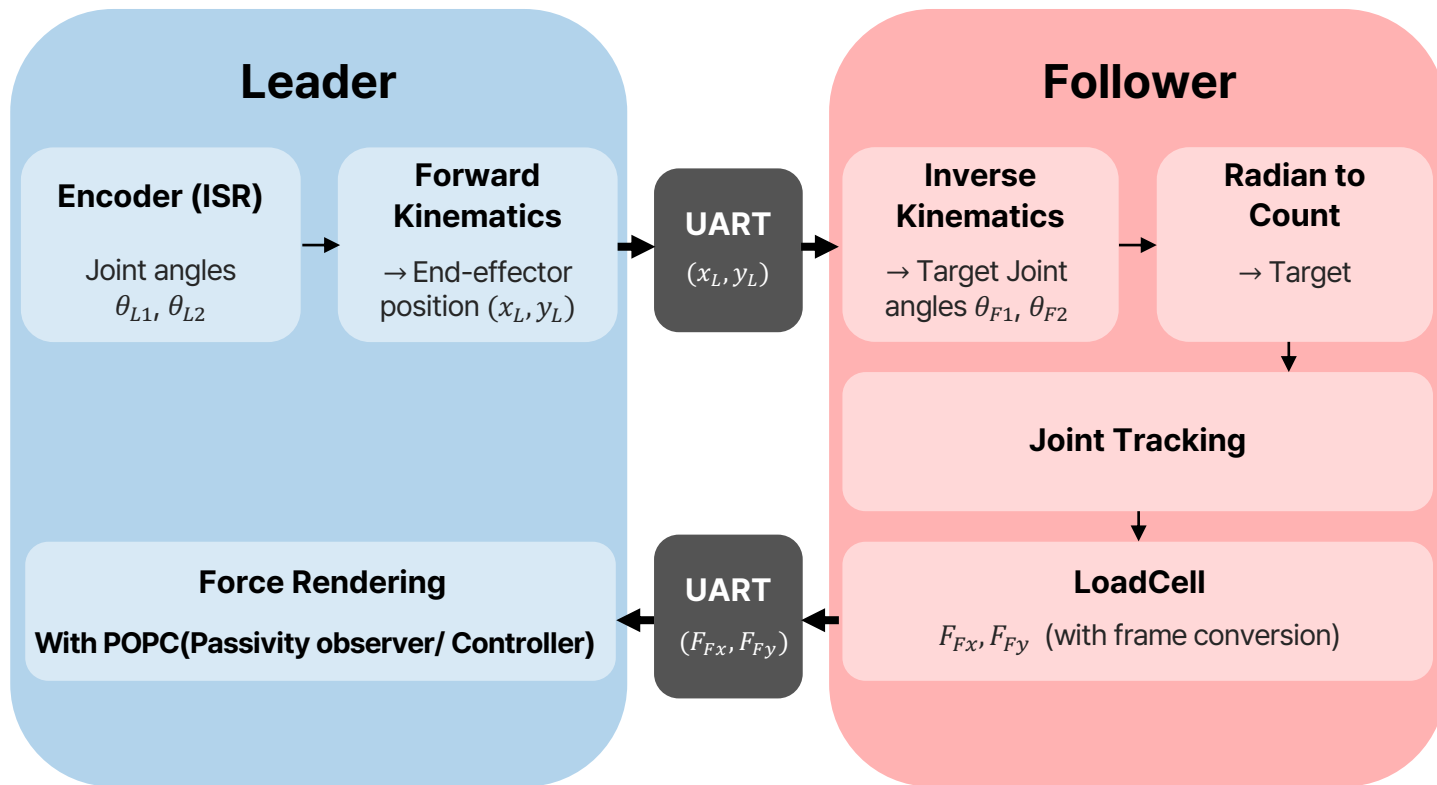
For the Teensy analog input:
 $f_c = 3.39\text{kHz}$, $\tau = 0.047\text{ms}$

Stable

For the load cell reference V:
 $f_c = 31.8\text{Hz}$, $\tau = 5\text{ms}$



C. In-depth improvement on Master-Slave Control Flow



Reliable Encoder Reading using ISR for **Stability**

Leader

Encoder (ISR)

Joint angles

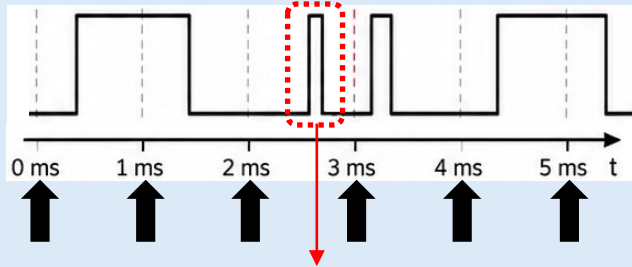
θ_{L1}, θ_{L2}

position

Force Rendering

Polling

In 1kHz:

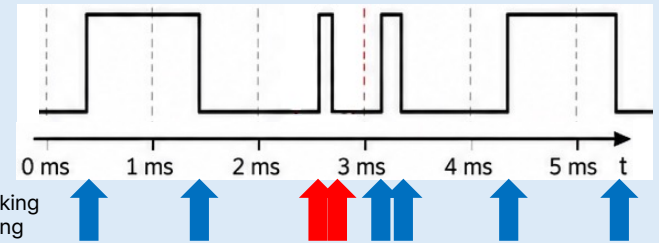


Possible missed counts

: Fast transitions between CPU checks can be missed.

vs.

ISR(Interrupt Service Routine)



Reduced risk of missed counts

: ISR captures signal changes immediately, reducing missed counts.

Packet Validation using CRC-16 for **Stability**

Leader

Encoder (ISR)

Joint angles
 θ_{L1}, θ_{L2}

Forward Kinematics

→ End-effector position (x_L, y_L)

UART
 (x_L, y_L)

Force Rendering

UART
 (F_{Fx}, F_{Fy})

Transmitter

UART

Receiver

Noise, timing error, baud-rate mismatch, or buffer issues can **corrupt the packet**

[Header] [Seq] [Data]

e.g., $F_x = 0.5N$

[Header] [Seq] [Corrupted Data]

e.g., $F_x = 50N$

→ **unexpected** motor command and **unstable motion**.

Safe UART communication with CRC-16

Tx

[Header] [Seq] [Correct Data] [CRC]

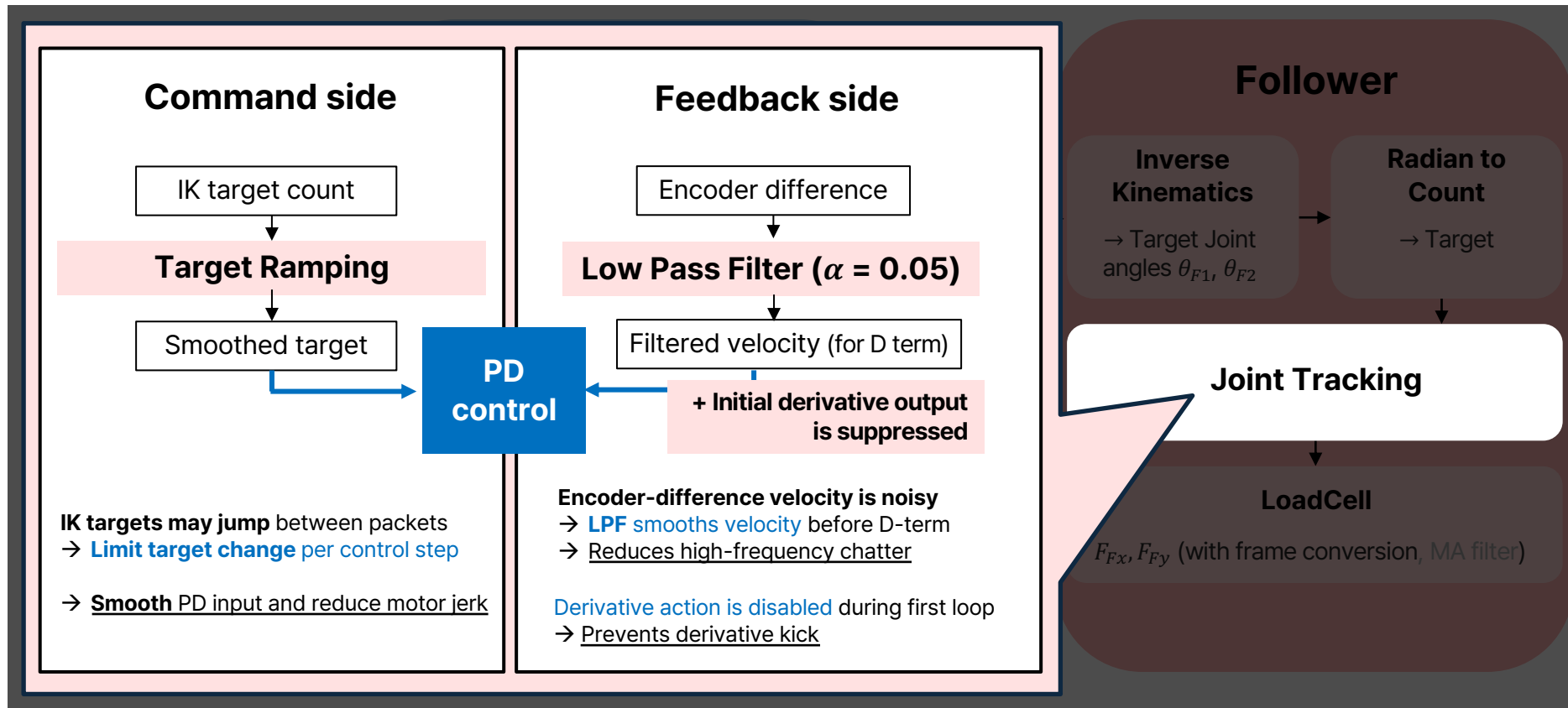
Accept

Rx

[Header] [Seq] [Corrupted Data] [CRC]

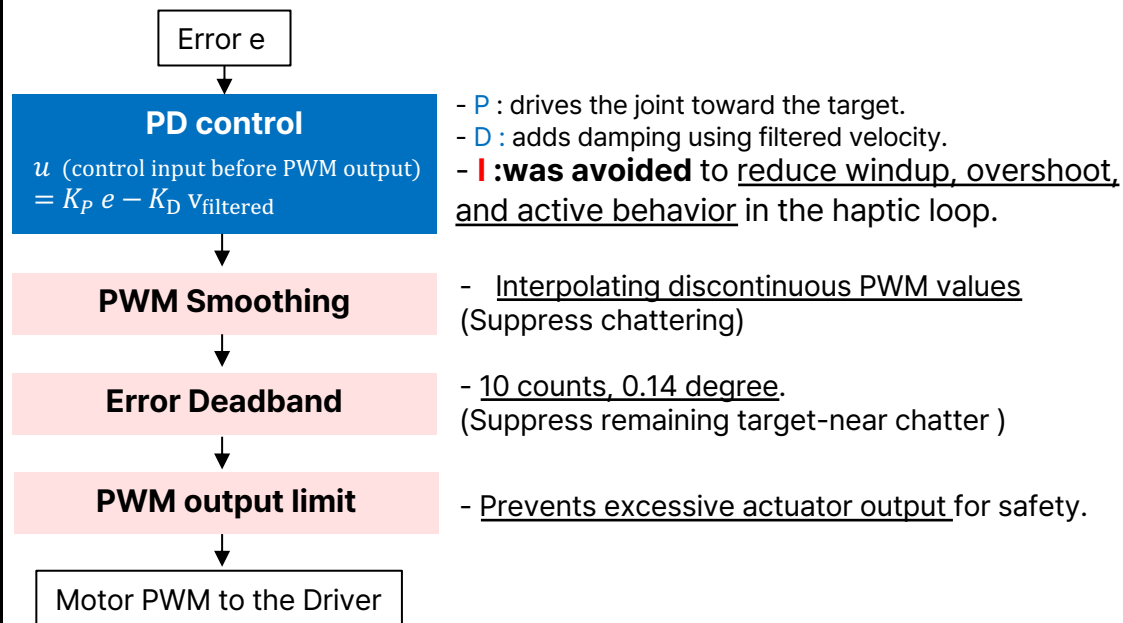
Reject

Input and Feedback Conditioning for **Stable** Joint Tracking

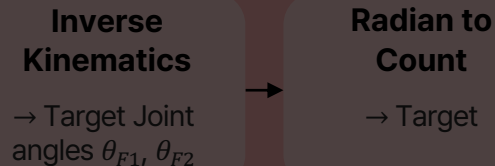


Output Conditioning for **Stable** and **Transparent** Joint Control

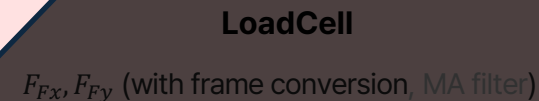
Output side:



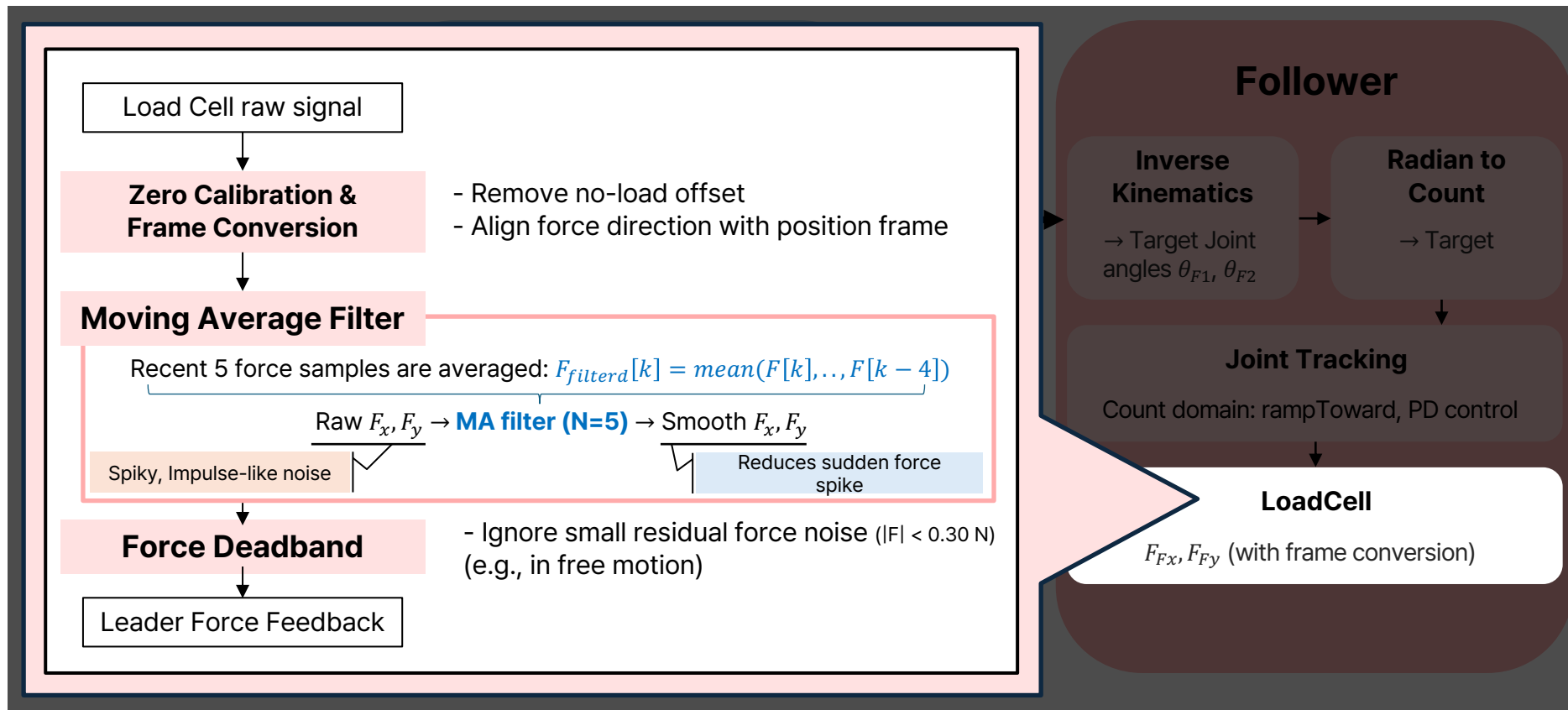
Follower



Joint Tracking



Force Signal Conditioning for **Stable** and **Transparent** Feedback



3. Improvements on the Teleoperation

Passivity Observer and Controller for **Stable** Feedback

Leader

Encoder (ISR)

Joint angles
 θ_{L1}, θ_{L2}

Forward Kinematics

→ End-effector position (x_L, y_L)

UA
 (x_L, y_L)

Force Rendering

With POPC(Passivity observer/ Controller)

Force feedback & Master velocity

- F_x, F_y from Slave

Passivity Observer

$E_{obs} \geq 0$: system is dissipating or strong passive energy

$E_{obs} < 0$: system may be generating active energy

→ *passivity violation risk*

Passivity Controller

→ If $E_{obs} < 0$, add damping

Jacobian transpose mapping

Compute the joint torque

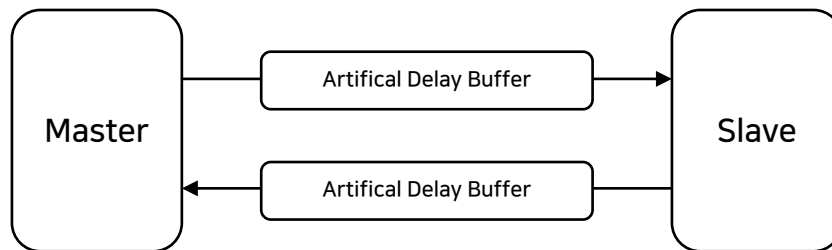
- Torque-to-PWM

Master motor

Passivity Observer and Controller for **Stable** Feedback



How can we evaluate the role of POPC?



Two-way artificial delay

C. Communication and latency minimization

- **Asynchronous Non-blocking Loops**
 - Both devices run a 1 kHz timer-based control loop.
 - Avoids `delay()` based blocking for more deterministic updates.
- **Localized Computation: Position-Force control**
 - **Master** computes handle position and force feedback.
 - **Slave** performs position tracking and load-cell force sensing.
- **Robust Data Exchange Structure**
 - Master sends position commands, mode flags, POPC state, delay state, and zero-force commands.
 - Slave returns measured position, joint tracking states, load-cell force, and status.
- **Delay/Passivity Evaluation**
 - Artificial forward/backward delay can be inserted.
 - **POPC** is applied to maintain stability under delay.

Fast local loop
→ Transparency, Stability

Stability,
through passivity control



Thank you

Hardware

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Minseop Kim, Seohee Kong, Sanghun Lee

Software

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